Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 00 (2026) 000–000

Procedia Computer Science

www.elsevier.com/locate/procedia

From Zero-Shot Transfer to Label-Efficient Adaptation: Evaluating Digital-Twin-Assisted Skeleton Action Recognition in Industrial Workcells

Ameur Gargouri[a,b], Mohamed Karray*[b], Mohamed Ksantini[a]

[a]*ATISP Lab, ENET'Com, University of Sfax, Sfax, Tunisia*
[b]*ESME Research Lab, Ivry-sur-Seine, France*

Abstract

In industrial settings, a recognition system must identify workers' actions across different workcells, yet collecting and annotating physical recordings for every worker, object configuration, and execution style is costly. A Digital Twin (DT) can provide labelled skeleton sequences in a controllable virtual source domain, but those sequences differ from physical target-domain recordings in posture, timing, object interaction, and workcell context. In this work, we study whether a recogniser trained in the source domain can transfer to unseen physical workers without target labels, and how much target supervision is needed when zero-shot transfer is insufficient. We study virtual-to-physical skeleton action-recognition transfer between the Industrial Human Action Recognition Dataset - Digital Twin (InHARD-DT) and Industrial Human Action Recognition Dataset (InHARD) datasets using a common Channel-wise Topology Refinement Graph Convolutional Network (CTR-GCN) backbone, four subject-disjoint target folds, and three random seeds. We compare source-only transfer with five unsupervised adaptation strategies. Across these strategies, Source-only achieved 40.59 ± 2.93% macro-F1-score (F1), compared with 39.34 ± 2.50% for Physics-Informed Cross-Modal Domain Adaptation (PI-CDA). With target labels, joint prototype fine-tuning from the digital-twin initialisation reaches 66.27 ± 3.46% macro-F1 using 50 labelled clips per class, versus 75.22 ± 4.04% for the target-supervised oracle.

© 2026 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)
Peer-review under responsibility of the scientific committee of the KES International.

Keywords: human-robot collaboration; digital twin; skeleton action recognition; sim-to-real transfer; few-shot learning; domain adaptation; industrial workcells

1. Introduction

Human activity recognition supports Human-Robot Collaboration (HRC). A robot sharing an assembly cell with a person must recognise the person's actions to select assistance, coordinate handovers, and maintain a safe separation. Skeleton representations are compact, relatively privacy-preserving, and capture spatial structure relevant to manipulation. Recent work on heterogeneous skeleton representations and higher-order topology identifies body structure and variation across skeletons as useful inductive biases for fine-grained actions [14, 1].

Industrial action recognition needs data covering different workers, workstations, object configurations, and execution styles. Collecting and labelling every combination in a physical cell is expensive and may interrupt production. A DT can generate task sequences in a virtual replica before deployment and expose otherwise rare actions. Recent

1877-0509 © 2026 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)
Peer-review under responsibility of the scientific committee of the KES International.

digital-twin systems also use synthetic-data pipelines [12]. Virtual recordings still differ from physical ones: workers adjust their posture around real objects, the workstation limits their reach, and the two settings produce different timing and co-articulation patterns, even when they share a kinematic hierarchy.

We test whether a recogniser trained on InHARD-DT transfers to InHARD, and whether digital-twin pretraining helps when only a small number of physical labels are available. Zero-shot Unsupervised Domain Adaptation (UDA) and few-shot supervised adaptation address different deployment settings. We use a fixed training protocol, subject-disjoint target test folds, and three random seeds; target labels do not inform model selection.

Our contributions are:

1. A reproducible real-data benchmark for virtual-to-physical skeleton action recognition using four subject-disjoint target folds and three seeds, with a common CTR-GCN backbone and fixed evaluation protocol, extending the earlier InHARD-DT digital-twin study [4].
2. A controlled comparison of source-only transfer, distribution alignment (Maximum Mean Discrepancy (MMD), Correlation Alignment (CORAL), and Domain-Adversarial Neural Network (DANN)), pseudo-label adaptation, PI-CDA, a target-supervised oracle, and joint source-target training.
3. A complete label-budget curve for the evaluated benchmark, covering 1, 5, 10, 20, and 50 labelled target clips per class across seven initialisation strategies and 12 runs per configuration.
4. An analysis showing that the evaluated zero-shot adaptation methods do not close the real-domain gap. Few-shot performance improves as the target-label budget grows, while the strongest tested strategy only modestly exceeds the few-shot source-only baseline beyond one shot.

2. Related Work

Recent studies address skeleton representation and recognition with limited labels. Other work examines sim-to-real transfer and physics-informed adaptation in industrial settings. We review these topics and digital-twin applications to frame our virtual-to-physical evaluation.

Wang et al. [14] studied skeleton observations with different joint dimensions and topologies, motivating representations that account for body structure. We use a canonical body19 representation and test transfer to previously unseen physical workers.

Zhang et al. [16] studied data-efficient skeleton action recognition with one-shot and limited data. Their work on extracting information from scarce labels relates to our target-label budget. We measure the value of digital-twin pretraining rather than propose a new aggregation module.

Chen et al. [1] proposed a dual-branch hypergraph model that combines channel-differential processing with dilated temporal modelling. Their approach uses higher-order skeletal relations. We keep the CTR-GCN recogniser fixed so that performance differences reflect transfer strategies rather than backbone capacity.

Liu et al. [8] reviewed preprocessing, representation, temporal modelling, and evaluation in 3D skeleton action recognition. Following this task-oriented approach, we report canonicalisation, subject-disjoint folds, and label budgets alongside classifier performance.

Rothert et al. [11] described the challenges of sim-to-real transfer for an industrial robotics task. Their case motivates evaluating models on the physical task rather than inferring deployment performance from simulation. We distinguish source-domain performance, zero-shot target transfer, and few-shot adaptation.

He et al. [5] proposed an interpretable, physics-informed method for cross-machine transfer diagnosis. Their study provides an industrial example of incorporating physical structure into adaptation. Our PI-CDA configuration applies a task-specific analogue by coupling feature alignment with skeleton-topology consistency. We evaluate it under worker and workcell shift.

Valentini et al. [13] developed a digital toolkit that combines Virtual Reality (VR) and Human-Machine Interface (HMI) prototypes for human-centred machine design, illustrating the use of digital tools in engineering. We assess digital-twin data by measuring how many physical labels are needed to reach a given target-domain performance.

Dallel et al. [4] introduced InHARD-DT as an auto-labelled digital-twin data generator and reported an Spatial-Temporal Graph Convolutional Network (ST-GCN) virtual-to-physical baseline. We use their work for dataset provenance and context, then evaluate transfer with a fixed CTR-GCN backbone, four subject-disjoint target folds, three

seeds, and explicit few-shot budgets. The published ST-GCN result is contextual because its protocol and backbone differ from ours; it is not a numerically paired result in our table.

These studies cover structured skeleton representations, data-efficient recognition, physical transfer, and digital-twin-assisted deployment. We examine these topics in one subject-disjoint benchmark and report both positive and negative results.

3. Methodology

Figure 1 traces the evaluation from the virtual InHARD-DT and physical InHARD datasets through skeleton harmonisation, transfer and target-label adaptation, to testing on held-out workers.

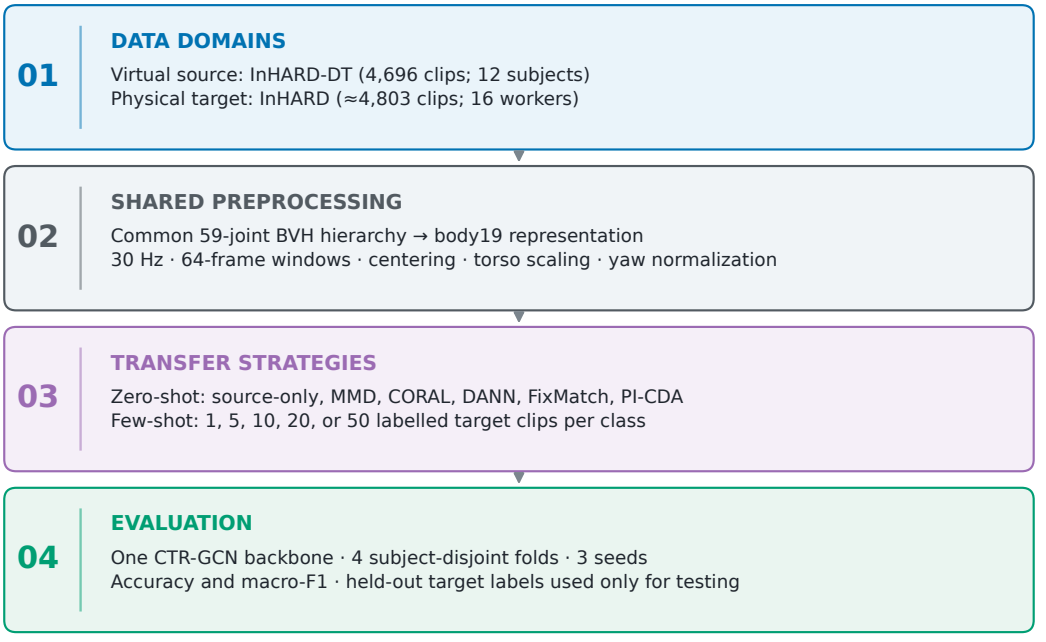


Fig. 1. Four-stage workflow from paired datasets through harmonization and transfer to held-out evaluation.

3.1. Problem Formulation

Let $\mathcal{S} = \{(x_i^s, y_i^s)\}$ be labelled skeleton clips from the digital-twin domain, and let $\mathcal{T} = \{x_j^t\}$ be unlabelled clips from the physical workcell. For zero-shot transfer, we learn f_θ from $\mathcal{S}$ and unlabelled $\mathcal{T}$, then evaluate it on held-out target subjects. For few-shot adaptation, we also use a support set $\mathcal{T}_k^\ell$ with k labelled clips per class. Test subjects remain disjoint from the source subjects and the target subjects used for adaptation.

3.2. InHARD-DT and InHARD

The InHARD-DT source domain contains 4,696 segmented clips from 12 subjects across 26 sessions. Participants wore a Perception Neuron suit in a virtual-reality replica of the workstation [2]. The InHARD physical target domain

contains approximately 4,803 clips from 16 workers recorded with the same Motion Capture (MoCap) system family [3]. We exclude "No action" clips. Both datasets use the same 59-joint Biovision Hierarchy (BVH) hierarchy and 13 meso-level action classes. The experiment measures transfer across physical context, worker population, execution style, and workstation interaction, rather than transfer between sensors. Figure 2 contains three original dataset images. The upper-left image shows a participant as a virtual avatar in the InHARD-DT workstation; the upper-right image shows four rendered views of that workstation with its original panel labels. The lower image is the InHARD montage, showing participants performing the assembly task alongside the corresponding skeleton view. We reproduce all three images unchanged under the Creative Commons Attribution 4.0 International license [2, 3]; the InHARD-DT study describes the virtual capture setup [4].

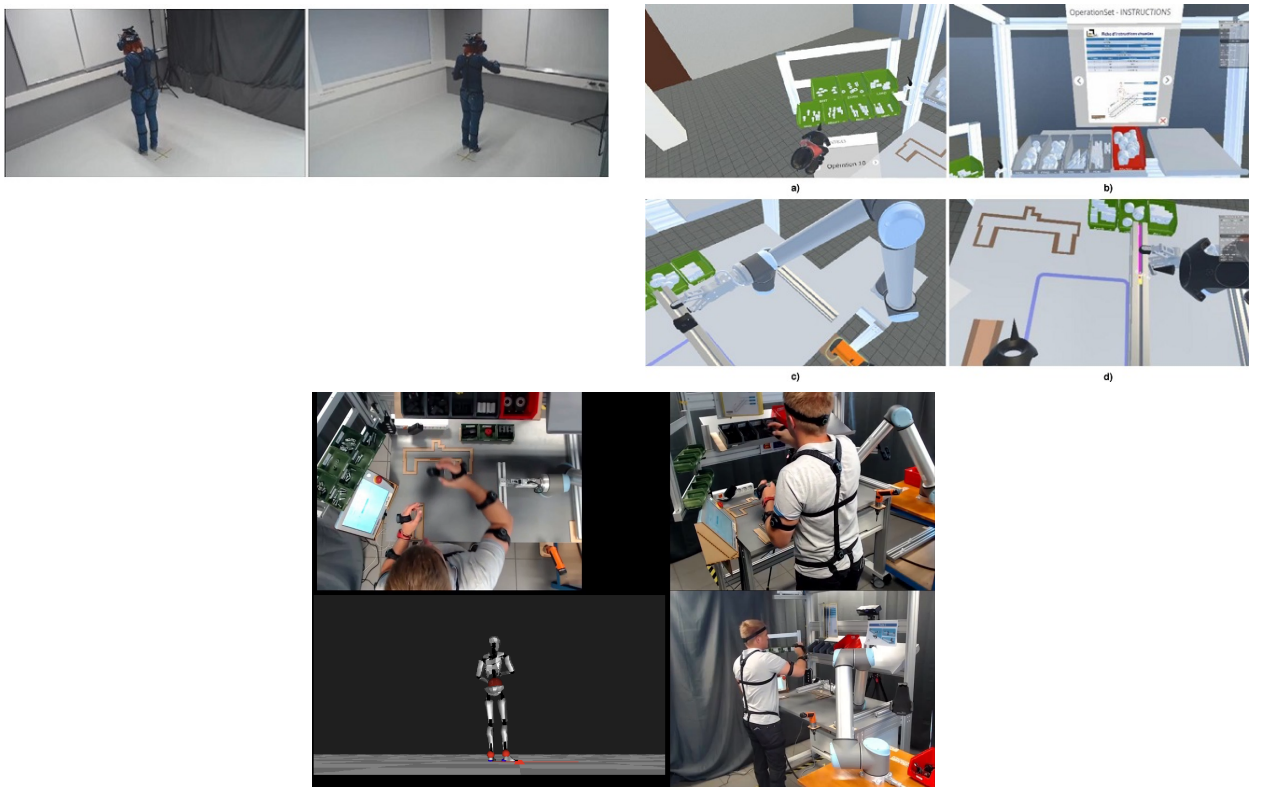


Fig. 2. Experimental setups for InHARD-DT and InHARD [2, 3, 4].

3.3. Preprocessing

We convert raw BVH sequences into fixed-length clips at 30 Hz and retain a 19-joint body representation comprising the torso, limbs, wrists, and hand landmarks used by the model. We canonicalise each clip by centring it on the first frame, scaling by torso size, and normalising its initial facing yaw. This reduces nuisance variation without using action labels or target-test information. Skeleton-based recognition requires consistent representations of body structure and temporal motion [8, 14]. Both domains use the same joint-coordinate representation and preprocessing pipeline.

3.4. Subject-disjoint folds

We divide the 16 InHARD workers into four fixed test folds, each with four workers. The remaining 12 workers in each fold form the target adaptation pool. We use all InHARD-DT subjects for the source domain and exclude held-out target workers from training; their labels are used only for final evaluation. Unless noted otherwise, reported

means and standard deviations cover four folds and three seeds. For Wilcoxon signed-rank comparisons, we pair seed-averaged per-subject macro-F1 values across the 16 target workers using the saved fold assignments. We report uncorrected pairwise p-values as exploratory results for this benchmark, not as equivalence tests across datasets. Subject-independent validation helps assess generalisation to unseen people, and recent human-activity-recognition comparisons examine how validation choices affect reported performance [10].

3.5. Common recogniser

All methods use the same CTR-GCN backbone, 64-frame windows, body19 input, 50 supervised epochs, batch size 64, and optimiser configuration recorded with the benchmark outputs. This control prevents methods from benefiting from a larger backbone or a different training budget. We train the source-only model with source labels and evaluate it on the target test fold without target labels. The DT in-domain result provides a diagnostic upper bound for cross-domain representation quality. The target oracle and source-plus-target model show performance when target supervision is available. Recent sensor-based activity-recognition work also uses graph-based domain adaptation, which motivates testing transfer methods with a fixed recogniser [15].

3.6. Algorithm Selection

We chose methods that test different explanations for transfer failure. MMD matches distributions with a kernel, CORAL aligns second-order statistics, and DANN learns adversarial domain invariance. FixMatch (FM) applies confidence-filtered pseudo-label consistency between weak and strong target views. Recent work analyses the generalisation of this semi-supervised approach [7]. Our PI-CDA configuration tests a project-specific physics-informed hypothesis by coupling feature alignment with skeleton-topology consistency. Together, the methods cover marginal and adversarial alignment, pseudo-labelling, and structure-aware regularisation.

The supervised controls answer three related questions. The source-only model measures transfer without target supervision; the target oracle measures performance when the target training pool is labelled; and source-plus-target training tests whether source data remains useful once target labels are available. The few-shot variants isolate the effect of the target-label budget across seven initialisations: scratch, source-only, PI-CDA, joint, FM, prototype, and prototype fine-tuning. All methods share the same CTR-GCN backbone, data representation, fold assignment, training budget, and final-model evaluation. Figure 1 shows the method selection and evaluation flow.

3.7. Few-shot adaptation

For each $k \in \{1, 5, 10, 20, 50\}$, the support set contains k labelled target clips per action class. We sample clips without replacement within each class from the 12-worker adaptation pool. The sampling seed is $1000s+10f+k$, where s is the experiment seed and f is the target fold. We save the selected clip identifiers with each result. We compare seven initialisations: scratch, source-only, PI-CDA, joint source-target training, joint training with FM, joint training with a prototype objective, and joint prototype fine-tuning. The final variant, `joint_proto_ft`, adapts the digital-twin-trained representation using labelled target clips and a prototype-aware objective. Recent work on cross-modal contrastive pre-training for few-shot skeleton action recognition provides context for this limited-label setting [9]. Support clips come only from the 12-worker adaptation pool; the four-worker test fold remains untouched.

3.8. Metrics and statistical protocol

We report Accuracy (ACC) and macro-F1. Macro-F1 is the primary metric because the 13 actions differ in difficulty and macro averaging weights each class equally. Recent work compares macro, micro, and support-weighted summaries in multi-label evaluation [6]. We compute means and standard deviations over 12 fold-seed runs. Paired comparisons use saved subject-level predictions and Wilcoxon signed-rank tests. Bootstrap intervals are reported with their seed coverage and do not replace the complete 12-run summaries. We do not use target test scores for early stopping or model and method selection.

4. Results

4.1. Zero-shot transfer

Table 1 compares the in-domain reference, zero-shot source transfer, five unsupervised adaptation methods, and two target-supervised controls. Macro-F1 is the primary comparison because it gives equal importance to each action class; the reported means and standard deviations cover four subject-disjoint folds and three seeds.

Table 1. InHARD-DT to InHARD transfer over four subject-disjoint folds and three seeds.

Method	ACC (%)	Macro-F1 (%)
DT in-domain	75.47 ± 3.54	73.32 ± 4.04
Source-only	50.16 ± 2.97	**40.59 ± 2.93**
DANN	46.71 ± 3.89	38.17 ± 3.00
CORAL	48.17 ± 4.00	38.88 ± 3.10
MMD	49.21 ± 3.18	39.66 ± 3.00
PI-CDA	48.35 ± 2.79	39.34 ± 2.50
FM	50.75 ± 2.91	40.54 ± 2.93
Target oracle	82.67 ± 2.38	75.22 ± 4.04
Source + target	**83.95 ± 1.98**	**75.89 ± 3.20**

This comparison evaluates recognition on unseen physical workers without target labels and tests whether unsupervised adaptation reduces the transfer gap. Each score covers four subject-disjoint folds, with four workers held out per fold, and three random seeds (12 fold-seed runs). Source-only transfer achieves 40.59 ± 2.93% macro-F1; DANN, CORAL, MMD, and PI-CDA achieve 38.17 ± 3.00%, 38.88 ± 3.10%, 39.66 ± 3.00%, and 39.34 ± 2.50%, respectively, while FM is effectively matched at 40.54 ± 2.93%. The PI-CDA result is 1.25 points below source-only and the paired difference is not significant ($p = 0.105$). We find no reliable zero-shot gain from the evaluated adaptation methods. The target oracle reaches 75.22 ± 4.04% and joint source-target training reaches 75.89 ± 3.20%, compared with 73.32 ± 4.04% for the digital-twin in-domain reference. The 34.63-point gap between source-only transfer and the target oracle points to generalisation across physical domains as the main challenge. In the available paired comparison, the oracle and source-plus-target results are statistically indistinguishable. Class-wise results show the same uneven pattern: source-only performs well on *Picking left* (77.1% F1) but poorly on *Consult sheets* (12.5%), *Take component* (18.8%), and *Take subsystem* (16.4%), for which the oracle reaches 30.6%, 74.5%, and 60.8%, respectively. Figure 3 compares methods by macro-F1; the dashed line marks the source-only result. The earlier ST-GCN study provides dataset context [4]. Its results are not directly comparable with this stricter CTR-GCN evaluation.

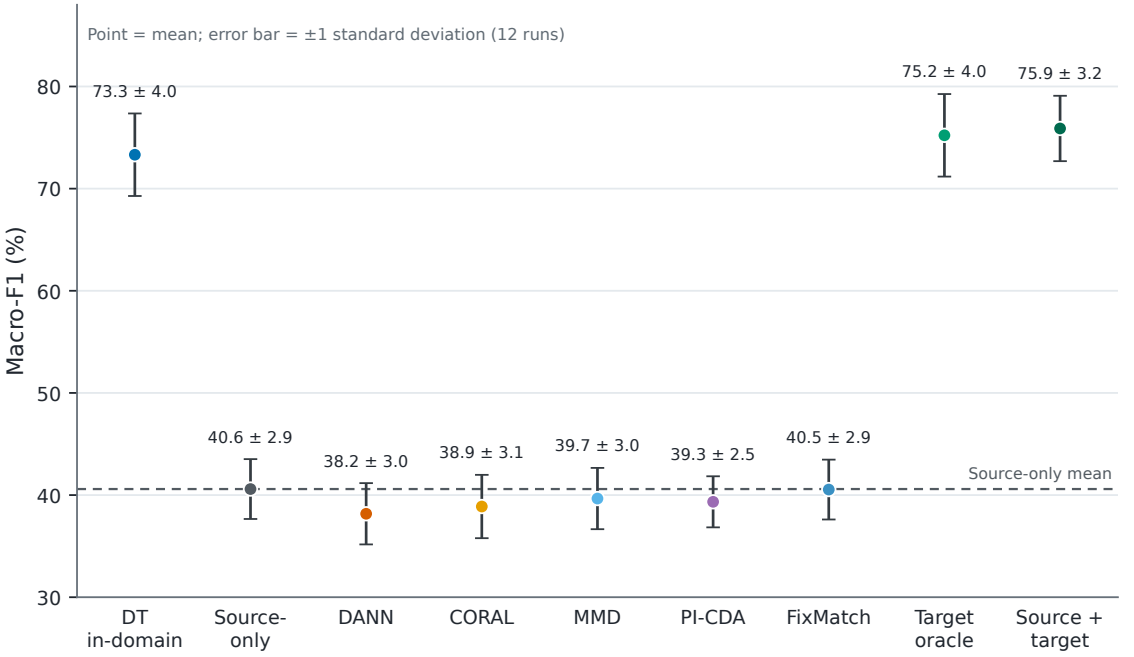


Fig. 3. Zero-shot virtual-to-physical transfer on the held-out physical domain.

4.2. *Effect of target-label budget*

Table 2 reports ACC and macro-F1 for joint prototype fine-tuning at five physical-label budgets, alongside the target oracle. Each value summarizes 12 fold-seed runs.

Table 2. Few-shot target adaptation with joint prototype fine-tuning, all values use 12 runs.

Target clips per class	ACC (%)	Macro-F1 (%)
1	50.35 ± 4.04	41.22 ± 4.38
5	57.91 ± 3.44	49.43 ± 3.74
10	61.99 ± 3.14	54.52 ± 3.35
20	67.17 ± 3.16	60.20 ± 3.96
50	74.11 ± 2.36	66.27 ± 3.46
Target oracle	82.67 ± 2.38	75.22 ± 4.04

This experiment measures recognition when physical target clips are labelled for adaptation. In each fold, we sample the support set from the 12-worker adaptation pool and evaluate on the four held-out workers. `joint_proto_ft` adapts the digital-twin-trained representation using the labelled clips. Its performance improves monotonically as the budget increases from 1 to 50 clips per class: macro-F1 is 41.22±4.38%, 49.43±3.74%, 54.52±3.35%, 60.20±3.96%, and 66.27 ± 3.46%, while ACC is 50.35 ± 4.04%, 57.91 ± 3.44%, 61.99 ± 3.14%, 67.17 ± 3.16%, and 74.11 ± 2.36%, respectively. Compared with zero-shot source-only macro-F1 (40.59%), the gains are 0.63, 8.84, 13.93, 19.61, and 25.68 points. At 50 clips per class, the method reaches about 74% of the 34.63-point gap between source-only transfer and the target oracle (75.22%). Target labels improve performance over unsupervised transfer. The curve does not establish whether digital-twin initialisation outperforms other ways of using those labels; the initialization-strategy comparison examines that question. Figure 4 plots both metrics against the target-label budget, with means and standard deviations over 12 fold-seed runs.

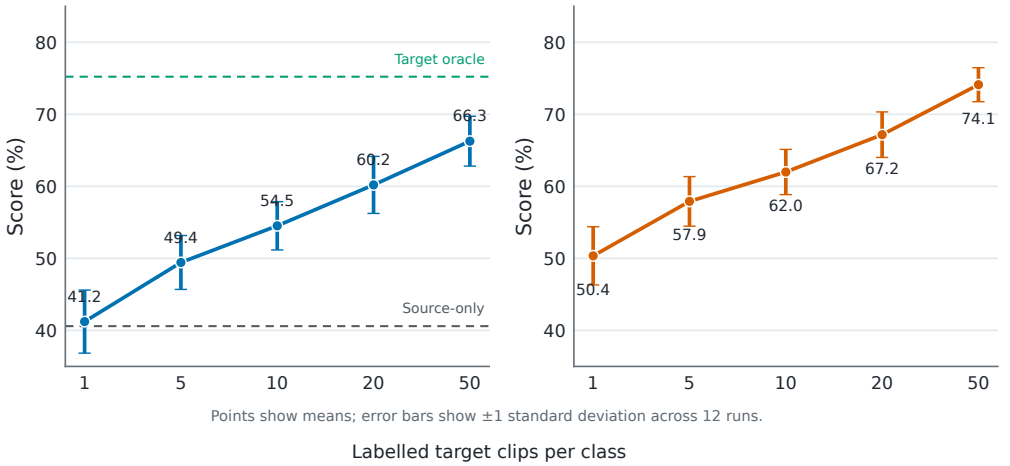


Fig. 4. Accuracy and macro-F1 for joint prototype fine-tuning.

4.3. Comparison of initialization strategies

This comparison tests whether performance depends on how virtual-source and physical-target data are combined, as well as on the number of target labels. We use the same support-set sampling and held-out-worker evaluation across five target-label budgets. Each plot compares seven strategies: training from scratch, source-only, PI-CDA, joint source-target training, joint training with FM, joint training with a prototype objective, and joint prototype fine-tuning. Horizontal position shows mean macro-F1; whiskers show one standard deviation over 12 fold-seed runs. Green identifies the numerical leader for a budget and does not indicate statistical significance.

With one labelled clip per class, joint training with a prototype objective has the highest mean macro-F1 (43.17 ± 4.08%), 13.89 points above few-shot source-only. The nearest joint configuration reaches 42.50%; the difference in means alone does not establish a significant advantage. Figure 5 shows the one-shot comparison.

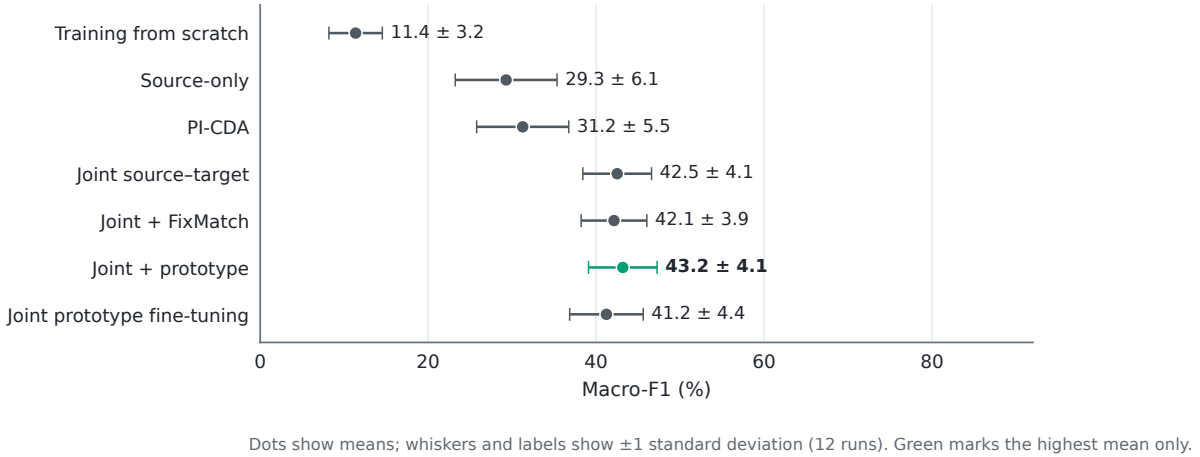
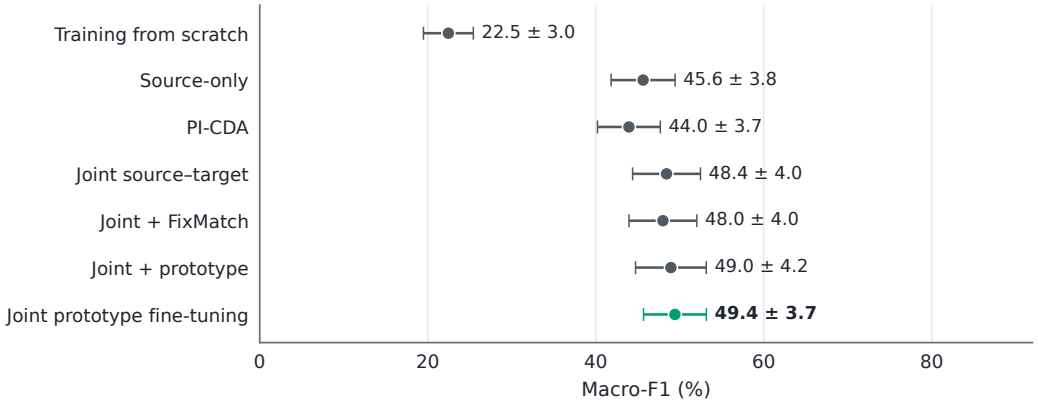


Fig. 5. Initialization strategies with 1 labelled target clip per class.

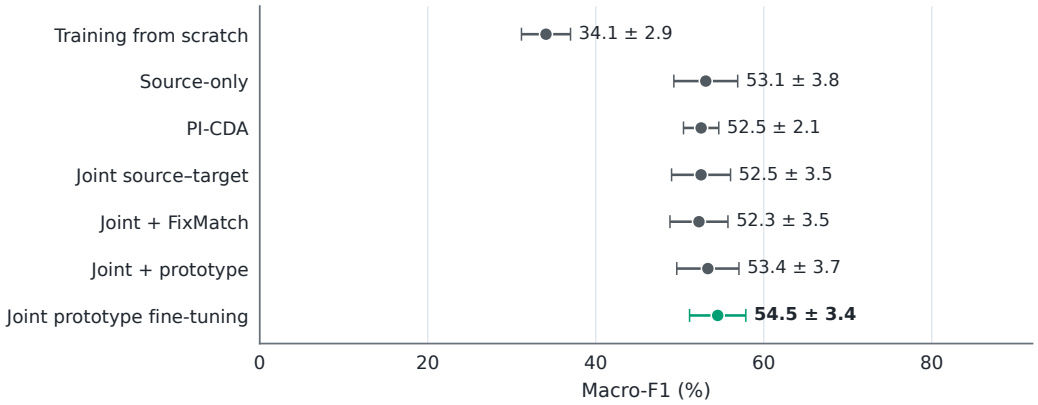
With 5 labelled clips per class, joint prototype fine-tuning reaches 49.43 ± 3.74% macro-F1, 3.79 points above few-shot source-only. Figure 6 compares the strategy means and their run-to-run variability at this budget.



Dots show means; whiskers and labels show ±1 standard deviation (12 runs). Green marks the highest mean only.

Fig. 6. Initialization strategies with 5 labelled target clips per class.

With 10 labelled clips per class, joint prototype fine-tuning reaches 54.52 ± 3.35% macro-F1, compared with 53.10% for few-shot source-only, a difference of 1.42 points. Figure 7 shows the seven strategies at this budget. The difference between the leading means is descriptive and does not establish statistical significance.



Dots show means; whiskers and labels show ±1 standard deviation (12 runs). Green marks the highest mean only.

Fig. 7. Initialization strategies with 10 labelled target clips per class.

With 20 labelled clips per class, joint prototype fine-tuning reaches mean macro-F1 of 60.20 ± 3.96%, 1.09 points above few-shot source-only (59.11%). Figure 8 shows this budget separately. The small difference between the leading means should be read in light of the run-to-run variability.

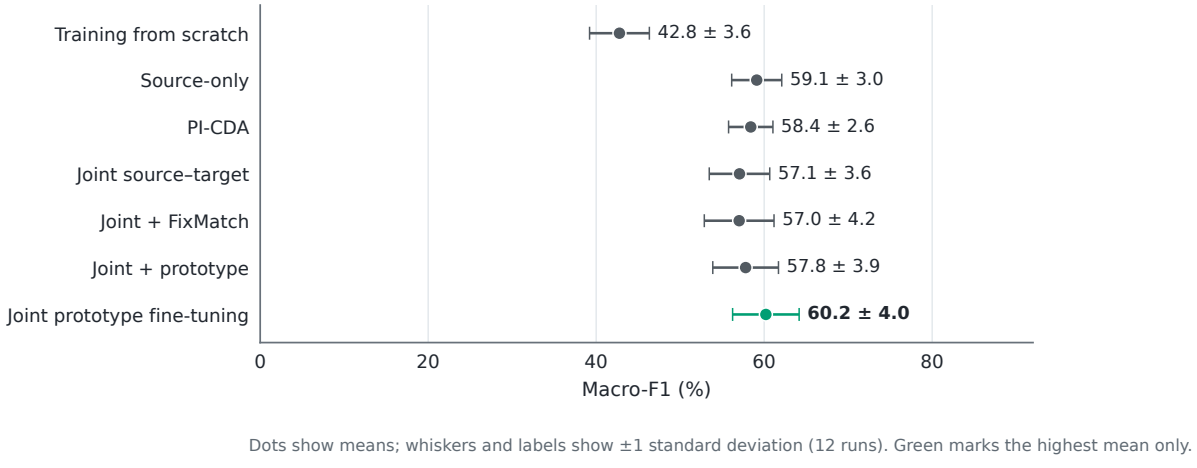


Fig. 8. Initialization strategies with 20 labelled target clips per class.

At 50 labelled clips per class, joint prototype fine-tuning reaches 66.27 ± 3.46% macro-F1, 1.28 points above few-shot source-only (64.99%). Figure 9 shows this budget separately. The leading strategy varies by budget, and its mean advantage over source-only is modest from 5 clips per class onward.

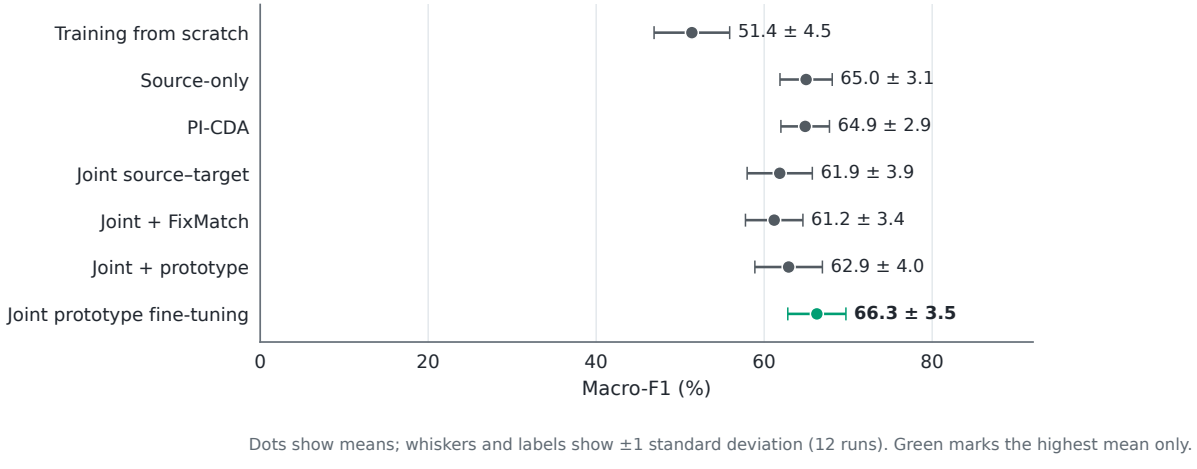


Fig. 9. Initialization strategies with 50 labelled target clips per class.

The experiments cover zero-shot transfer to held-out physical workers, the effect of adding physical labels, and the performance of alternative initialization and adaptation strategies. The evaluated unsupervised methods do not reliably improve on source-only transfer, so these results do not show that the digital twin alone closes the virtual-to-physical domain gap. Performance rises with the target-label budget. Prototype-based adaptation performs competitively with one labelled clip per class, but its numerical advantage over few-shot source-only is modest at larger budgets and does not establish statistical significance. These results characterize zero-shot transfer and label-efficient adaptation under a controlled, subject-disjoint protocol. They do not establish a general solution or a replacement for physical data. The findings apply to the tested InHARD-DT-to-InHARD pair, shared CTR-GCN backbone, and evaluation protocol; testing on additional datasets and workcells is needed to assess their broader relevance.

5. Discussion

5.1. What the digital twin does and does not provide

The in-domain DT result does not predict physical deployment performance: macro-F1 falls from 73.32% in-domain to 40.59% under source-only transfer. The shared BVH hierarchy and canonical skeleton reduce representational nuisance, but the domain gap remains. PI-CDA does not improve zero-shot transfer in this benchmark, which marks a limit of the evaluated configuration.

Target supervision improves few-shot performance, but the results do not show that digital-twin pretraining accounts for the full gain. Training from scratch is weakest, while the prototype-based strategy has the highest mean at each budget. Its margin over few-shot source-only is largest with one clip per class and modest from 5 to 50 clips. Digital-twin data may provide a useful starting point when labels are scarce, although additional physical labels remain important and no tested initialization consistently produces a large advantage over the other few-shot strategies.

5.2. Implications for deployment

For a new workcell, one possible procedure is to pretrain on the digital twin, collect a small, class-balanced calibration set from the target worker population, and adapt without using held-out workers' labels. The required budget depends on risk tolerance and class costs. In this benchmark, 20 to 50 clips per class outperform one-shot calibration by a substantial margin. The target oracle remains an upper bound, not an operational requirement.

5.3. Threats to validity

The benchmark has four limitations. First, it uses one source-target pair and one primary backbone. Generalisation to other robots, sensors, and action taxonomies remains untested; our findings characterize this transfer setting and do not establish how all digital twins or adaptation methods behave. Second, the few-shot support-set sampling and class distribution are specific to this protocol. Third, we evaluate recognition on held-out workers, not closed-loop robot assistance, so improved F1 does not establish safety. Fourth, the statistical comparison file contains bootstrap intervals with limited seed coverage. We base the primary claims on complete 12-run summaries and paired fold-subject analyses. Further evaluation should include multiple workcells and risk-sensitive measures.

6. Conclusion

Collecting labelled recordings from every worker and workcell is costly, which makes digital-twin data attractive for industrial action recognition. However, virtual recordings may not represent how workers move and interact with equipment in a physical cell. We examined this transfer problem using InHARD-DT as the source and InHARD as the target. With a shared CTR-GCN backbone, we compared source-only transfer and unsupervised adaptation under four subject-disjoint folds and three random seeds, then evaluated few-shot adaptation with target labels while keeping test workers held out.

The results show that the tested unsupervised methods did not reliably improve zero-shot performance: source-only transfer reached 40.59% macro-F1, compared with 39.34% for PI-CDA. With 50 labelled target clips per class, joint prototype fine-tuning reached 66.27%, below the 75.22% target oracle. The gains as the label budget increased show that digital-twin pretraining can provide a starting point for adaptation, while physical labels remain important. Further work should test this finding across more workcells, worker groups, and sensing conditions, and evaluate whether recognition gains support calibrated, risk-sensitive robot assistance.

References

[1] Chen, D., She, K., Wu, P., Chen, M., Li, C., 2025. Dual-branch differential channel hypergraph convolutional network for human skeleton-based action recognition. PLOS ONE 20, e0332066. doi:10.1371/journal.pone.0332066.

[2] Dallel, M., Havard, V., Baudry, D., Dupuis, Y., 2023a. Inhard-dt – industrial human action recognition dataset – digital twin. URL: https://zenodo.org/records/7644247, doi:10.5281/zenodo.7644247.

[3] Dallel, M., Havard, V., Baudry, D., Savatier, X., 2020. Inhard – industrial human action recognition dataset in the context of industrial collaborative robotics. URL: https://zenodo.org/records/4003541, doi:10.5281/zenodo.4003541.

[4] Dallel, M., Havard, V., Dupuis, Y., Baudry, D., 2023b. Digital twin of an industrial workstation: A novel method of an auto-labeled data generator using virtual reality for human action recognition in the context of human–robot collaboration. Engineering Applications of Artificial Intelligence 118, 105655. doi:10.1016/j.engappai.2022.105655.

[5] He, C., Shi, H., Liu, X., Li, J., 2024. Interpretable physics-informed domain adaptation paradigm for cross-machine transfer diagnosis. Knowledge-Based Systems doi:10.1016/j.knosys.2024.111499.

[6] Hinojosa Lee, M.C., Braet, J., Springael, J., 2024. Performance metrics for multilabel emotion classification: Comparing micro, macro, and weighted f1-scores. Applied Sciences 14, 9863. doi:10.3390/app14219863.

[7] Li, J., Pan, J., Tan, V.Y.F., Toh, K.C., Zhou, P., 2024. Towards understanding why fixmatch generalizes better than supervised learning. arXiv preprint arXiv:2410.11206 doi:10.48550/arXiv.2410.11206.

[8] Liu, M., Liu, H., Hu, Q., Ren, B., Yuan, J., Lin, J., Wen, J., 2025. 3d skeleton-based action recognition: A review. arXiv preprint arXiv:2506.00915 doi:10.48550/arXiv.2506.00915.

[9] Lu, M., Yang, S., Lu, X., Liu, J., 2024. Cross-modal contrastive pre-training for few-shot skeleton action recognition. IEEE Transactions on Circuits and Systems for Video Technology 34, 9798–9807. doi:10.1109/TCSVT.2024.3402952.

[10] Mekruksavanich, S., Hnoohom, N., Phaphan, W., Jitpattanakul, A., 2025. A comparative study of validation methods for sensor-based human activity recognition using deep learning models, in: 2025 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECTI DAMT & NCON), pp. 686–691. doi:10.1109/ECTIDAMTNCON64748.2025.10962074.

[11] Rothert, J.J., Lang, S., Seidel, M., Hanses, M., 2024. Sim-to-real transfer for a robotics task: Challenges and lessons learned, in: 2024 IEEE 29th International Conference on Emerging Technologies and Factory Automation (ETFA). doi:10.1109/ETFA61755.2024.10711073.

[12] Salierno, G., Leonardi, L., Cabri, G., 2024. A big data architecture for digital twin creation of railway signals based on synthetic data. IEEE Open Journal of Intelligent Transportation Systems 5, 342–359. doi:10.1109/OJITS.2024.3412820.

[13] Valentini, L., Mengoni, M., Peruzzini, M., 2025. Digital toolkit for human-centered machine design: Development and testing of an innovative system integrating virtual reality and hmi digital prototypes. International Journal of Advanced Manufacturing Technology 136, 415–432. doi:10.1007/s00170-025-16943-4.

[14] Wang, H., Ma, X., Kuang, J., Gui, J., 2025. Heterogeneous skeleton-based action representation learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 19154–19164. URL: https://openaccess.thecvf.com/content/CVPR2025/html/Wang_Heterogeneous_Skeleton-Based_Action_Representation_Learning_CVPR_2025_paper.html, doi:10.1109/CVPR52734.2025.01784.

[15] Yang, J., Liao, T., Zhao, J., Yan, Y., Huang, Y.C., Zhao, Z., Xiong, J., Liu, C., 2024. Domain adaptation for sensor-based human activity recognition with a graph convolutional network. Mathematics 12, 556. doi:10.3390/math12040556.

[16] Zhang, Z., Cai, W., Liu, Q., Wang, Y., 2025. Skeletonx: Data-efficient skeleton-based action recognition via cross-sample feature aggregation. IEEE Transactions on Multimedia doi:10.1109/TMM.2025.3618561.